

Data Visualization Document

Project Title: HealthFirst Care Performance Dashboard

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Tools Used: Google Looker Studio, Microsoft Excel

Datasets: Cleaned_Appointments, Cleaned_Feedback, Cleaned_Resources

1. Overview

This document describes the design process, datasets used, analytical approach, and key business insights derived from the interactive dashboard developed for HealthFirst Care. The primary objective of this dashboard is to support management decision-making by visualizing critical healthcare operational metrics, including patient wait times, resource utilization, and patient satisfaction levels.

The dashboard is designed to enable stakeholders to:

- Monitor operational performance.
- Identify resource bottlenecks and inefficiencies.
- Understand patient experience trends.
- Support data-driven decisions for service quality improvement.

2. Datasets and Variables Used

Three cleaned datasets were utilized in this project. These datasets were preprocessed to ensure consistency, accuracy, and analytical readiness.

2.1 Cleaned_Appointments

Purpose: Analyze patient wait time trends and appointment flow.

Key Variables:

- AppointmentID: Unique identifier for each appointment.
- PatientID: Unique identifier for patients.
- DoctorID: Identifier for attending physicians.
- Department: Clinical department handling the appointment.
- Appointment Date: Date of scheduled visit.
- Time: Appointment time.
- Status: Appointment completion status.

Derived Metrics:

- Average Patient Wait Time (minutes)
- Monthly Wait Time Trend

2.2 Cleaned_Feedback

Purpose: Measure patient satisfaction and service quality.

Key Variables:

- FeedbackID: Unique identifier for feedback records.
- PatientID: Unique identifier for patients.

- Department: Clinical department handling the patient.
- Feedback Score: Numerical feedback rating.
- Comments: Qualitative feedback.
- Wait Time (in min): Recorded patient waiting duration.
- Satisfaction Level: Categorical variable (High / Low). If feedback score less than 4 it categorize as “Low”, and vice versa.

Derived Metrics:

- Percentage of High vs Low Satisfaction
- Department-level satisfaction distribution

2.3 Cleaned_Resources

Purpose: Analyze operational efficiency and resource allocation.

Key Variables:

- ResourceID: Unique identifier for medical equipment or staff.
- Resource Type: Type of resource (equipment, room, staff category, etc.).
- Department: Assigned department.
- Usage Hours: Total time resource was used.
- Date: Date of resource usage.
- Availability: Resource availability status.

Derived Metrics:

- Resource Utilization Rate (%)
- Resource Usage Distribution by Department

3. Dashboard Design Process

The dashboard design followed a structured analytical and visualization methodology to ensure clarity, usability, and executive relevance.

3.1 Requirement Analysis

Based on stakeholder needs and healthcare operational priorities, three primary analytical dimensions were identified:

1. **Patient Experience:** Waiting time trends and satisfaction levels.
2. **Operational Efficiency:** Resource utilization and distribution.
3. **Departmental Performance:** Identification of high-load and underutilized units.

These dimensions guided the selection of KPIs and visualization techniques.

3.2 Visualization Strategy

The dashboard layout was divided into three logical sections:

A. Executive KPI Summary (Header Section)

- Average Patient Wait Time
- Average Resource Utilization Rate

Purpose: Provide instant situational awareness for decision-makers.

B. Patient Experience & Trend Monitoring (Top Section)

1. Line Chart – Patient Wait Time Trends

- X-Axis: Appointment Date (monthly)
- Y-Axis: Average Wait Time (minutes)
- Filter: Department

Purpose: Identify fluctuations, seasonal patterns, and operational stress periods.

2. Pie Chart – Patient Satisfaction Levels

- Metrics: High vs Low satisfaction
- Values: Percentage distribution
- Filter: Department

Purpose: Evaluate overall or per department service quality perception.

C. Resource Utilization Analysis (Bottom Section)

1. Bar Chart – Resource Utilization Rate by Department

- X-Axis: Department
- Y-Axis: Utilization Rate (%)

Purpose: Compare workload distribution across departments.

2. Treemap – Resource Usage Distribution

- Size: Count of Resource use based on Resource Type
- Color: Total Usage Hours
- Filter: Resource Type

Purpose: Identify high-demand departments and potential bottlenecks.

Note: Due to native visualization limitations in Looker Studio, a treemap visualization was implemented as a functional alternative to a traditional matrix heatmap. The treemap effectively represents both resource volume and intensity of usage through proportional sizing and color encoding.

3.3 Interactivity Design

To enhance analytical flexibility, interactive filters were implemented:

- Department filter
- Resource Type filter
- Date range filter

These filters allow stakeholders to conduct focused exploratory analysis and scenario evaluation.

4. Key Findings and Business Implications

4.1 Patient Wait Time Trends

Finding:

- The average patient wait time remains relatively stable at approximately 44.6 minutes throughout the observed period.

Business Implication:

- Although stable, the wait time remains relatively high for healthcare service standards, indicating opportunities for process optimization and capacity balancing.

4.2 Patient Satisfaction Levels

Finding:

- High satisfaction accounts for 79.8% of feedback responses.
- Low satisfaction accounts for 20.2%.

Business Implication:

- Overall service perception is positive.
- However, the significant proportion of low satisfaction suggests a need for targeted service improvements, particularly in departments with higher wait times or resource strain.

4.3 Resource Utilization by Department

Finding:

- Pediatric and Cardiology departments exhibit the highest utilization rates.
- Orthopedics and Oncology display comparatively lower utilization levels.

Business Implication:

- Resource distribution is imbalanced.
- Reallocation of staff, equipment, or operational capacity from underutilized departments to high-demand departments can improve overall efficiency and reduce patient wait times.

4.4 Resource Usage Distribution (Treemap Analysis)

Finding:

- Pediatric and General Medicine departments show the largest resource footprints and highest usage intensity.
- Orthopedics and Cardiology demonstrate comparatively lower demand.

Business Implication:

- Indicates potential operational bottlenecks in Pediatric and General Medicine.
- Supports strategic workforce planning and predictive staffing models.

5. Strategic Recommendations

Based on the analytical findings, the following recommendations are proposed:

1. Resource Rebalancing

Redistribute available staff and equipment capacity from underutilized departments to high-load units.

2. Predictive Capacity Planning

Use historical trends to forecast patient volumes and dynamically adjust staffing schedules.

3. Service Quality Monitoring

Implement SLA-based KPIs such as percentage of patients waiting longer than 30 minutes.

4. Operational Process Optimization

Review appointment scheduling, triage workflows, and patient flow design.

6. Conclusion

The HealthFirst Care Performance Dashboard provides an integrated and interactive view of patient experience and operational performance. By consolidating appointment data, patient feedback, and resource utilization metrics, the dashboard enables healthcare managers to identify inefficiencies, monitor service quality, and implement targeted improvement strategies.

This analytical framework supports data-driven decision-making and contributes to improved healthcare service delivery, operational efficiency, and patient satisfaction.